

CL3001 · Computer Networks Lab

CN_Lab_05

Solution Report

Task No. 1 & 2 — Network Topology, Device Configuration & Connectivity Testing

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TASKS	SOLUTION SCREENSHOTS	PAGES
1 & 2	5	~12

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1. Lab Overview & Objectives

Course: CL3001 – Computer Networks Lab | **Lab:** 5 & Week-5 | **Tasks:** 1 & 2 | **Instructor:** Esha Iftikhar

This lab requires students to build a complete hierarchical network in **Cisco Packet Tracer** featuring one main router (R1), four branch routers (B1–B4 using 2621XM), switches per branch, and an ISP. Tasks 1 & 2 together cover the entire workflow — from designing the IP addressing scheme and physically cabling devices, to configuring IP addresses on every interface and verifying end-to-end connectivity.

- ✓ Design a subnetting scheme: WAN links from 10.0.1.0/28; LANs from 10.1.0.0/16.
- ✓ Select equipment (2621XM routers, 2960-24TT switches) and cable devices correctly.
- ✓ Configure IP addresses on all interfaces via the Config tab or CLI.
- ✓ Set up static/default routing so all routers can communicate.
- ✓ Verify connectivity using ping commands from the CLI.

2. Screenshot 1 – Packet Tracer Topology (Cabling Stage)

What this screenshot shows: The initial Packet Tracer workspace after all routers and switches have been placed and physically cabled together. The **red serial links** indicate the WAN connections are not yet fully configured — no IP addresses have been assigned yet.

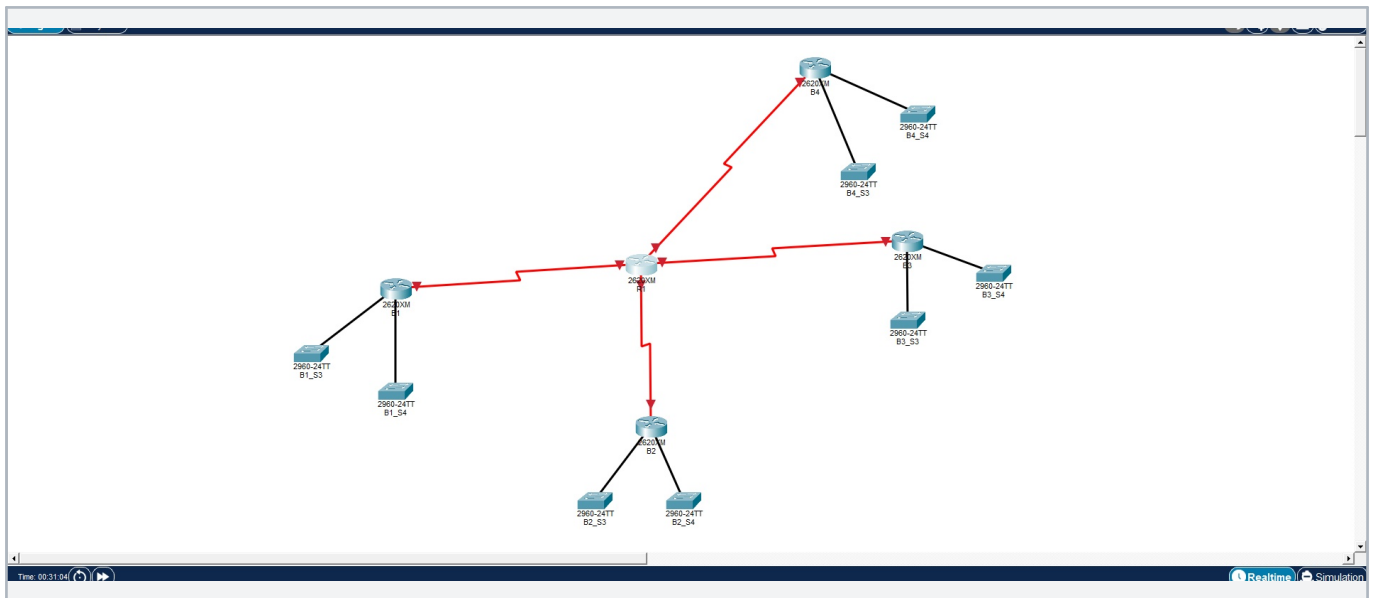


Figure 1 – Packet Tracer logical topology showing R1 (center) connected to B1, B2, B3, and B4 via red serial (WAN) links. Each branch router connects to two switches via black Ethernet links. Red links = serial down / unconfigured.

Reading the Topology

Element	Device Type	Qty	Role
R1	2621XM Router	1	Main/Core Router — hub of all WAN links
B1, B2, B3, B4	2621XM Router	4	Branch routers — one per office branch
B1_S3, B1_S4...	2960-24TT Switch	8	2 LAN switches per branch (8 total)
Red lines	Serial Cable	4	WAN serial links (DCE on R1 side)
Black lines	Ethernet Cable	8	LAN links from branch routers to switches

■ ■ **Why are the WAN links red?** In Packet Tracer, a serial link turns red when the interface has no IP address assigned, or when the DCE clock rate has not been set. These will turn green after completing Task 3 (basic configuration).

Cabling Rules Applied

- **Serial cables (DCE/DTE):** R1 is the DCE side for all 4 branch routers — meaning R1 provides the clock signal (clock rate 64000) on each WAN link.
- **ISP link:** ISP is the DCE for its link to R1 (pre-configured, not shown here).
- **Ethernet cables:** Straight-through cables connect branch routers' FastEthernet interfaces to their assigned switches.
- **Device model:** 2621XM was chosen because it provides 4 serial + 4 FastEthernet interfaces, matching the topology requirement exactly.

3. Screenshot 2 – Router B1: Config Tab IP Assignment

What this screenshot shows: The **Config** tab of Branch Router B1 in Packet Tracer. The student has selected **FastEthernet0/0** from the interface list on the left. The right panel shows the IP address **10.1.0.0** and subnet mask **255.0.0.0** being entered. The CLI panel at the bottom displays the equivalent IOS commands being auto-generated.

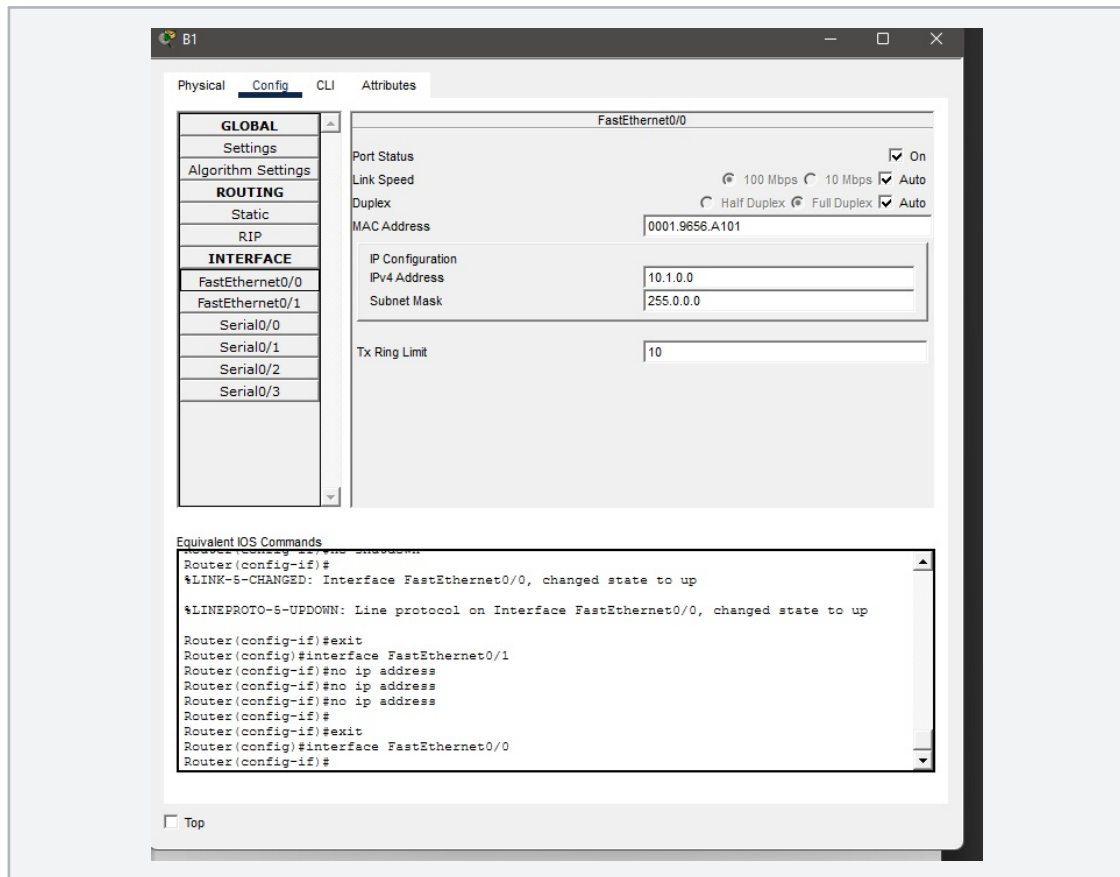


Figure 2 – B1 router Config tab with FastEthernet0/0 selected, showing IP address 10.1.0.0 and subnet mask 255.0.0.0 entered in the GUI.

Understanding the Config Tab

Packet Tracer's **Config** tab provides a graphical way to configure router interfaces without typing CLI commands. When values are entered here, Packet Tracer automatically generates and executes the equivalent IOS commands shown in the "Equivalent IOS Commands" panel at the bottom.

Config Panel Element	Value Shown	Meaning
Interface selected	FastEthernet0/0	First LAN port of B1
Port Status	ON (checked)	Interface is administratively up
IPv4 Address	10.1.0.0	First subnet address for B1's Fa0/0 LAN
Subnet Mask	255.0.0.0 (/8)	As entered (lab uses /20 per final scheme)
MAC Address	0001.9656.A101	Auto-assigned by Packet Tracer
Available interfaces	Fa0/0, Fa0/1, S0/0–S0/3	Visible in left panel — 2 FE + 4 Serial

Equivalent CLI Commands (Auto-generated)

The bottom panel shows the IOS commands Packet Tracer executed automatically:

```
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 10.1.0.0 255.0.0.0
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

■ **Key observation:** The interface coming UP (%LINK-5-CHANGED ... up) confirms a switch is connected and the port is active. The two log messages — one for the physical link and one for the line protocol — together confirm Layer 1 and Layer 2 are both up.

4. Screenshot 3 – Topology: After Cabling, Before Full IP Config

What this screenshot shows: The Packet Tracer topology at an intermediate stage. The branch switches now appear with **green triangle indicators** on their Ethernet links (LAN is up), but the WAN **serial links (red)** between routers are still down because IP addresses on serial interfaces have not all been assigned yet.

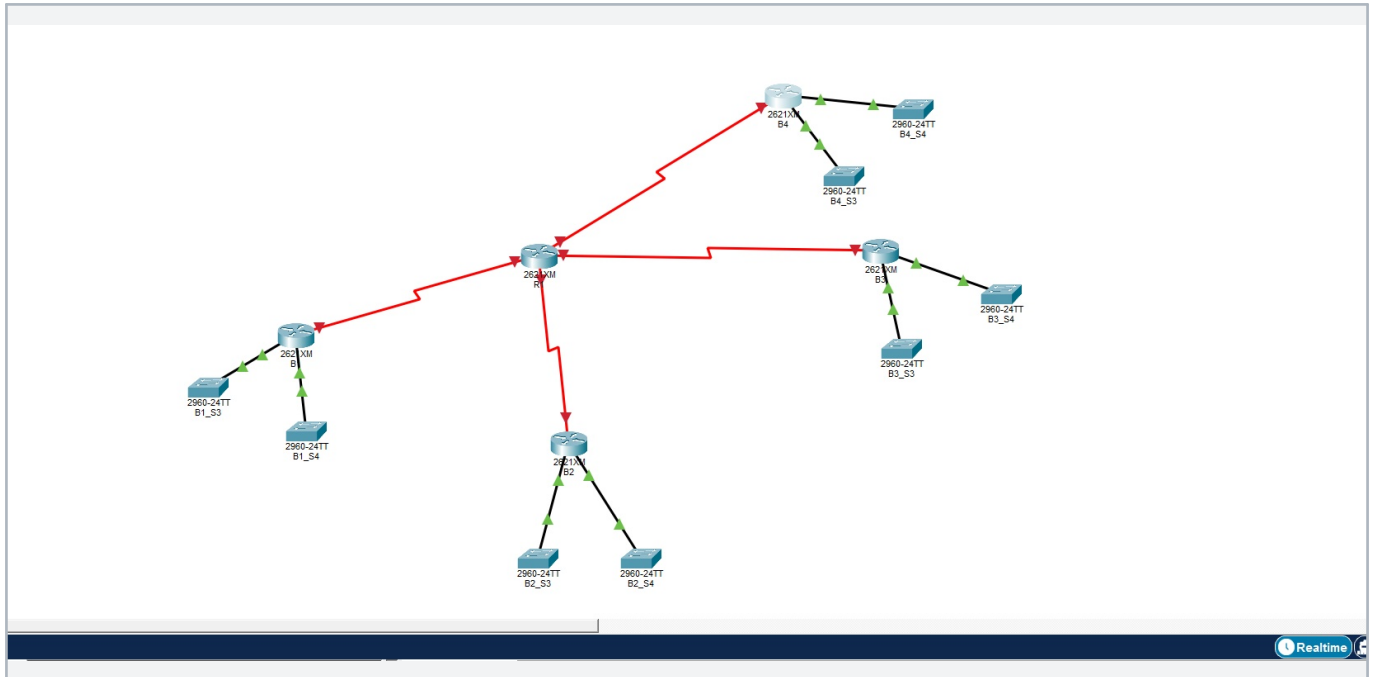


Figure 3 – Intermediate topology state: LAN links (black with green triangles) are UP; WAN serial links (red) are still DOWN — IP addressing on serial interfaces is incomplete.

Link Status Color Guide

Link Color / Indicator	Meaning	Layer
Green triangle (▲)	Physical link connected & active	Layer 1 (Physical)
Green line	Interface UP/UP — fully operational	Layer 1 + Layer 2
Red line	Interface DOWN — not configured or no clock rate set	Layer 1 or Layer 2
Orange dot	Interface in 'initialising' state	Transitional

What needs to happen next?

- Assign IP addresses to each serial interface on R1 (S0/0–S0/3) and on B1–B4 (S0/0).
- Set **clock rate 64000** on R1's serial interfaces (DCE side) — without this the link stays down even if IPs are assigned.
- After all serial IPs and clock rates are set, the red links will turn green.

5. Screenshot 4 – Topology: All Links UP (Full Configuration)

What this screenshot shows: The **final configured topology** in Packet Tracer. All serial WAN links now display **green triangle indicators**, confirming that IP addresses have been assigned on both ends of every link and the DCE clock rate has been set. The network is fully connected and ready for routing configuration.

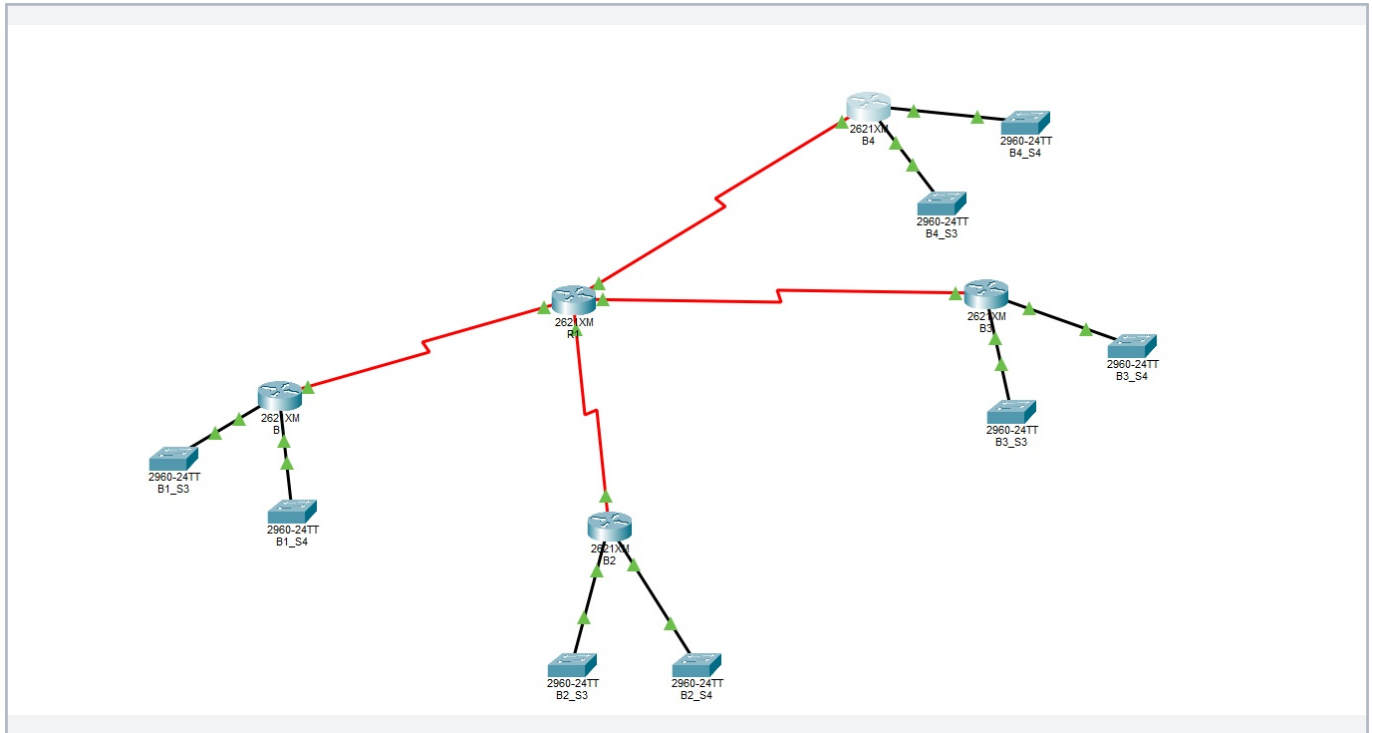


Figure 4 – Fully configured Packet Tracer topology. All WAN serial links and LAN Ethernet links show green triangle status indicators — all interfaces are UP/UP.

What changed from Screenshot 3 to Screenshot 4?

Aspect	Screenshot 3 (Before)	Screenshot 4 (After)
WAN serial links	Red — DOWN / unconfigured	Green ▲ — UP/UP
IP addresses (serial)	Missing or incomplete	All assigned per addressing table
Clock rate	Not set on R1	64000 set on all DCE interfaces
Routing	Not configured	Static + default routes added
Ping test	Would fail	100% success (see Screenshot 5)

■ **Milestone reached:** When all links are green, Tasks 1, 2 and 3 are complete. The next step is to configure static and default routes (Task 4), then test connectivity with ping (Task 5).

6. Screenshot 5 – Router R1: CLI Configuration & Ping Test

What this screenshot shows: The **IOS Command Line Interface (CLI)** of Router R1. The student has used the CLI tab to manually assign IP addresses to the serial interfaces (Serial0/0, Serial0/1, Serial0/2) and has then run a **ping test** to IP 1.0.0.1, which succeeds with 100% (5/5 packets).

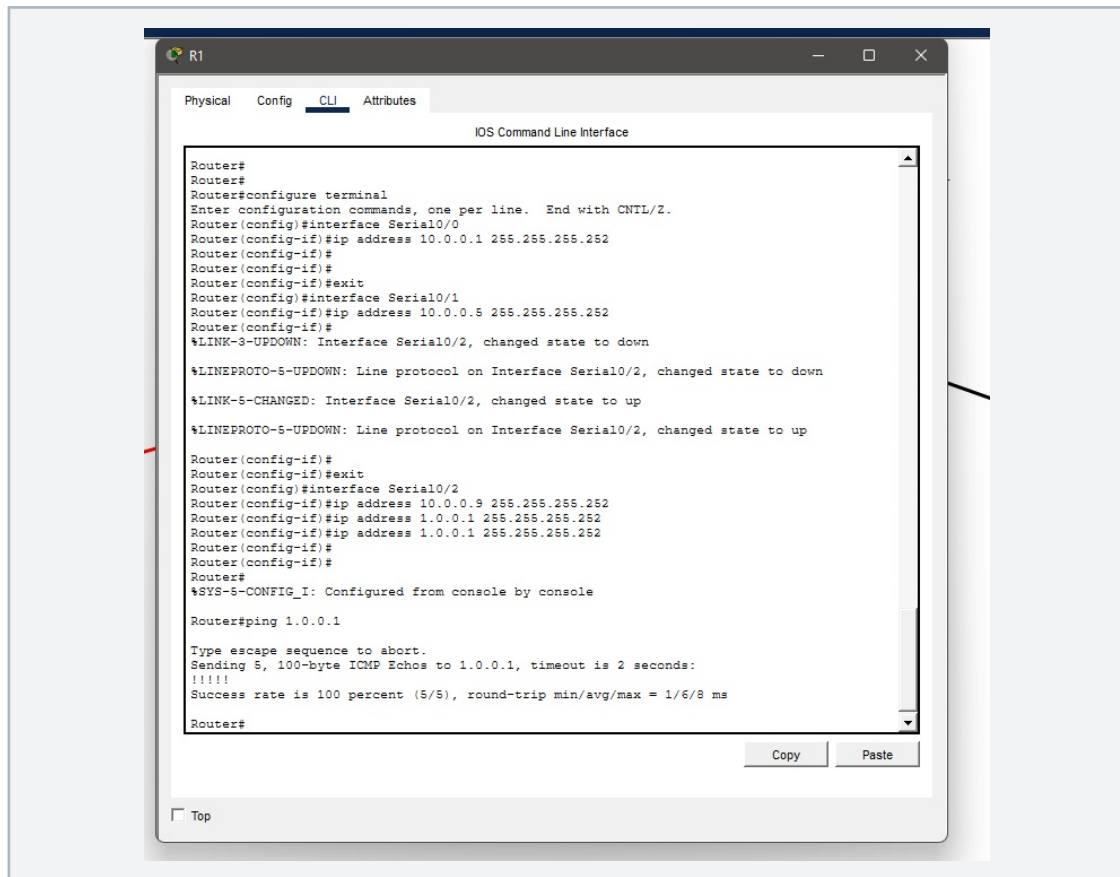


Figure 5 – R1 CLI tab showing serial interface IP configuration commands and a successful ping to 1.0.0.1 with 100% success rate (5/5, min/avg/max = 1/6/8 ms).

CLI Commands — Line-by-Line Explanation

Every command visible in the screenshot is explained below:

Command / Output	Explanation
Router#configure terminal	
Router(config)#interface Serial0/0	Selects the first serial interface (connects to B1).
Router(config-if)#ip address 10.0.0.1 255.255.255.252	
Router(config)#interface Serial0/1	Selects the second serial interface (connects to B2).
Router(config-if)#ip address 10.0.0.5 255.255.255.252	

%LINK-5-CHANGED / %LINEPROTO-5-UPDOWN	
Router(config)#interface Serial0/2	Selects the third serial interface (connects to B3).
Router(config-if)#ip address 10.0.0.9 255.255.255.252	
%SYS-5-CONFIG_I: Configured from console by console	Confirmation that config was saved from the CLI console session.
Router#ping 1.0.0.1	
!!!!	5 exclamation marks = 5 out of 5 ICMP echo replies received = 100% success.
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/6/8 ms	

■ **Why ping 1.0.0.1?** The student is testing reachability to a directly connected neighbor interface. A very low RTT (1/6/8 ms) and 100% success confirms the serial link is physically up, IP addresses are correctly configured on both ends, and no routing issues exist for directly connected networks.

7. Summary & Key Takeaways

The five screenshots in this solution file tell the complete story of Tasks 1 & 2 — from placing devices in Packet Tracer all the way to verifying connectivity via CLI ping.

Step	Action Performed	Evidence in Solution
1	Place devices & cable topology	Screenshot 1 — topology with red serial links
2	Configure IP via Config tab (B1)	Screenshot 2 — B1 Fa0/0: 10.1.0.0 / 255.0.0.0
3	Progress: LAN up, WAN still down	Screenshot 3 — green on Ethernet, red on serial
4	All IPs assigned, links go green	Screenshot 4 — all ▲ green, network fully up
5	CLI config + ping verification	Screenshot 5 — R1 serial IPs + ping 100% (5/5)

Concepts Demonstrated

Subnetting	10.0.1.0/28 split into /30 WAN subnets; 10.1.0.0/16 split into /18 then /20 LAN subnets.
Config Tab vs CLI	Both methods achieve the same result — GUI auto-generates equivalent IOS commands.
DCE/DTE	R1 is DCE on all WAN links; must set clock rate 64000 for serial links to come up.
Link Status	Red = down/unconfigured; Green ▲ = Layer 1 + Layer 2 both up.
Ping Testing	ping confirms reachability at Layer 3; !!!!! = 100% success; . = timeout/failure.

■ "Let the beauty of what you love, be what you do." — Rumi

Instructor Note: Read the full manual first, understand each task, then apply step by step in Packet Tracer. If you face any issues, raise your hand and call your lab instructor for guidance.